Beyond Mean Curvature: Lower-Tail Routing Structure in Controlled Hierarchical Networks

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Abstract

In prior control-law simulations, lower-tail Ollivier-Ricci curvature tracked transport optima more closely than mean curvature, suggesting that the lower curvature tail may mark a special routing structure in hierarchical networks. The Curvature Adaptation Hypothesis (CAH) control-law extension showed that the relevant operating point is generally an intermediate, condition-dependent optimum and that lower-tail curvature q_{10} is a more informative geometric correlate of that optimum than mean curvature alone. Here, we test the stronger possibility that the lower curvature tail is not merely a correlate of efficient transport, but a genuine load-bearing routing substructure.

Using repeated-seed transport simulations on balanced hierarchical graphs with sparse distal shortcuts, we compare targeted and random shortcut placement across multiple edge budgets. We show that lower-tail edges consistently bear disproportionate transport load, confirming that the lower curvature tail is a real routing structure rather than a descriptive artifact. However, lower-tail depth alone does not explain system performance. Across repeated seeds, targeted placement increases total tail participation most clearly in the sparse regime, whereas random placement more often produces greater per-edge tail burden concentration and higher tail exploitation intensity. These results indicate that the relevant control variable is not lower-tail depth in isolation, but the regime-dependent organization of the lower curvature tail across recruitment and burden concentration.

Taken together, the findings refine the geometric-control picture suggested in previous work. The transport optimum is better understood not as a function of average curvature or even lower-tail depth alone, but as a tradeoff between distinct modes of lower-tail organization. This reframes the lower tail from a scalar clue into an organized routing object and supports a broader view of hierarchical transport as governed by structured lower-tail recruitment rather than uniform curvature reduction.

1 Introduction

Hierarchical transport systems face a persistent geometric problem (Krioukov et al., 2010; Boguñá et al., 2010). As structure deepens, efficient routing requires the system to reduce traversal cost and congestion without paying unsustainable maintenance penalties for indiscriminate shortcut access. In the broader Curvature Adaptation Hypothesis (CAH), this problem is framed geometrically and thermodynamically: biological and synthetic systems may actively regulate effective curvature in order to reduce the transport burdens imposed by flat, Euclidean organization. Dynamic Curvature Adaptation (Pender, 2026a) argued that targeted gating can unlock a negative-curvature regime in which effective graph distances collapse, reducing the number of intermediate relay operations required for large-scale integration and thereby mitigating the practical burden imposed by irreversible information processing (Landauer, 1961). In that framework, geometry is not a static background property of the network, but a dynamic resource governed by control.

Our CAH control-law extension (Pender, 2026b) sharpened this picture by showing that the relevant operating point is not maximal shortcut permissiveness, maximal densification, or maximal negative curvature. Instead, a synthetic hierarchical transport model exhibited a robust intermediate optimum in a composite objective J that jointly balanced traversal cost, congestion, maintenance burden, and integration. Crucially, the most informative geometric correlate of this optimum was not mean curvature, but the lower tail of the curvature distribution, q_{10}. The optimum occurred at intermediate lower-tail curvature rather than at either the flattest or most extremely negative values, implying that efficient transport depends on preserving a selectively negative routing substructure rather than simply "hyperbolizing" the graph globally.

That result raised a sharper mechanistic question. If lower-tail curvature tracks the optimum more reliably than mean curvature, what exactly is the lower tail doing? One possible interpretation is that q_{10} functions as a scalar sufficient statistic: the lower-tail state becomes more informative as it deepens, and the optimum is therefore governed by the depth of the tail itself. But the CAH control-law extension already pointed toward a more complicated picture. The control parameter did not merely flatten or hyperbolize the network as a whole; it nonlinearly reshaped the most negatively curved routing substructure, and the best-performing states avoided both low-control bottlenecking and high-control shortcut saturation (Pender, 2026b).

The present study investigates that substructure directly. Rather than asking whether lower-tail curvature is merely correlated with efficient transport, we ask whether it marks a genuine routing scaffold within hierarchical graphs (Goh et al., 2001; Serrano et al., 2009). More specifically, we test whether sparse distal shortcuts placed in targeted versus random configurations organize the lower curvature tail in distinct ways, and whether those organizational differences help explain why lower-tail curvature outperforms mean curvature as a signal of the optimum. Our aim is therefore not to establish a blanket theorem of lower-tail optimality, but to determine what kind of transport object the lower tail actually is.

The central hypothesis of this paper is that the lower curvature tail forms a genuine load-bearing routing substructure, but that its contribution to global performance depends on how traffic is distributed across that tail rather than on tail depth alone (Arenas et al., 2001; Barthélemy, 2011). We distinguish at least two partially separable dimensions of lower-tail organization. The first is total tail participation, meaning how much of the system's total traffic runs through lower-tail edges. The second is per-edge tail burden concentration, meaning how intensely the participating tail edges are exploited relative to the rest of the graph. Under this framing, targeted sparse placement need not universally outperform random placement in the global objective. Instead, different interventions may organize the lower tail differently, broadening its recruitment or concentrating its burden, with global performance depending on the tradeoff between these modes.

This framing positions the paper as a mechanistic refinement of our CAH control-law extension rather than a departure from it. Dynamic Curvature Adaptation established that hierarchical transport systems can actively enter a negative-curvature regime under control (Pender, 2026a), and the extension established that the useful operating point is intermediate and is tracked better by lower-tail curvature than by mean curvature (Pender, 2026b). The present work takes the next step by asking whether q_{10} is best understood not as a magic scalar, but as a hint toward a richer routing object whose organization must itself be analyzed.

2 Hypothesis and Conceptual Framing

The CAH control-law extension showed that lower-tail curvature tracks transport optima more reliably than mean curvature (Pender, 2026b), implying that the most negatively curved portion of the network contains information that average geometry smooths away (Ni et al., 2015; Farooq

et al., 2019). However, that result alone does not establish what kind of object the lower tail is. One possibility is that lower-tail depth itself is the governing variable: deeper q_{10} would then directly imply better routing. Another possibility is more structured: the lower curvature tail may mark a genuine routing scaffold whose usefulness depends not simply on how negative it becomes, but on how traffic is distributed across it. The present study is designed to distinguish between these interpretations.

Our central hypothesis is therefore not merely that the lower curvature tail matters, but that it matters in a specifically organizational way. We hypothesize that the lower curvature tail in hierarchical transport networks forms a genuine load-bearing routing substructure, but that its contribution to global performance depends on how traffic is distributed across that tail rather than on tail depth alone (Borgatti and Everett, 2000; Yan et al., 2006). Under this framing, the relevant control variable is not lower-tail depth in isolation, but the regime-dependent organization of the lower tail across at least two partially separable dimensions: total tail participation and per-edge tail burden concentration. In practical terms, this means that two conditions with similar q_{10} values may nevertheless differ meaningfully in performance if one recruits the tail more broadly and the other exploits a smaller subset of tail edges more intensely.

This reframing matters because it sharpens the interpretation of the earlier CAH control-law result (Pender, 2026b). If lower-tail curvature is genuinely more informative than mean curvature, then the key explanatory object is unlikely to be a purely scalar quantity. Instead, q_{10} may be functioning as the first visible signature of a richer routing structure that must itself be analyzed. The present paper therefore treats lower-tail curvature not as the final answer, but as the clue that motivates a more explicit investigation of lower-tail organization.

2.1 Why Mean Curvature is Insufficient

Mean curvature is a natural first summary of graph geometry, but it is too coarse to explain the transport behavior observed in the CAH control-law simulations (Pender, 2026b). By averaging across all edges, mean curvature obscures the subset of routes that bear disproportionate transport burden under hierarchical constraints. In the earlier control-law setting, the optimum was not tracked most clearly by the graph-wide mean, but by the lower tail of the curvature distribution. This already implied that the most informative geometric signal was being carried by a restricted subset of edges rather than by the graph as a whole.

Conceptually, this is exactly what one would expect if efficient routing in hierarchical networks depends on a small number of structurally special pathways. Hierarchical graphs are not burdened uniformly (Barthélemy, 2004). Their transport cost is shaped disproportionately by bottlenecks, distal alternatives, and the availability of selective branch-bridging routes (Girvan and Newman, 2002). Any scalar average taken over the whole graph risks smoothing away the part of the network that actually determines routing efficiency. Mean curvature may still track broad geometric trends, but it cannot by itself isolate the substructure where transport burden is concentrated (Sandhu et al., 2015; Weber et al., 2017).

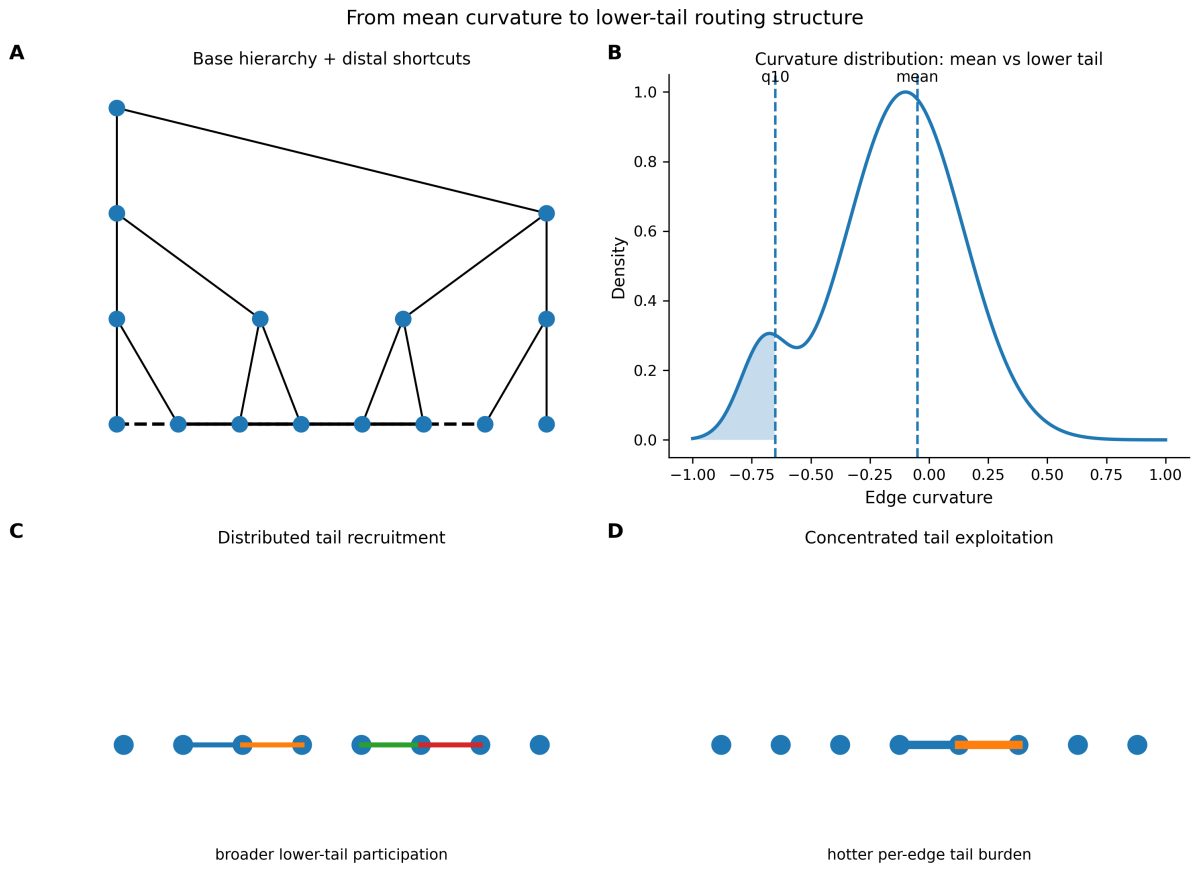

Figure 1: **From mean curvature to lower-tail routing structure.(A)** Schematic hierarchical transport graph with sparse distal shortcuts, illustrating the class of balanced-tree networks studied here. **(B)** Conceptual curvature distribution showing why mean curvature can obscure routing-relevant structure carried by the lower tail, summarized by q_{10}. **(C)** Distributed tail recruitment, in which lower-tail edges are broadly recruited into the transport substrate. **(D)** Concentrated tail exploitation, in which a smaller subset of lower-tail edges bears disproportionately intense per-edge burden. The figure introduces the paper's central conceptual shift: the relevant object is not average curvature alone, but the organization of the lower-curvature tail across participation and burden concentration.

The insufficiency of mean curvature therefore motivates the shift in emphasis that underlies the present study. If the optimum is better tracked by the lower tail than by the mean, then the key question is no longer whether geometry matters in general, but which part of the geometry is doing the work. This is the conceptual opening through which the lower-tail routing hypothesis emerges.

2.2 Lower-Tail Curvature as a Candidate Routing Structure

The lower curvature tail is a natural candidate for a routing-relevant substructure because it identifies the most negatively curved portion of the graph, precisely where hierarchical detours, branch-spanning shortcuts, and transport-critical alternatives are most likely to appear (Topping et al., 2022; Jonckheere et al., 2011). In the CAH control-law framework (Pender, 2026b), the optimum did not occur at the flattest geometry or at the most extreme negative curvature,

but at an intermediate lower-tail state. That pattern suggests that the lower tail is not simply a proxy for "more hyperbolicity," but a marker of how selectively negative routing structure is arranged within the graph.

The present study takes that suggestion literally. Rather than treating q_{10} as a scalar summary to be correlated against the objective, we treat the edges that fall within the lower tail as a candidate routing object in their own right. If those edges systematically bear disproportionate transport load, then the lower tail is not merely descriptive. It is a real routing substrate (Estrada, 2023). But if that is true, then the next question is not whether the tail exists, but how it is organized. A lower-tail structure may be useful because it recruits many tail edges into the routing burden, or because it concentrates burden intensely onto a smaller subset of tail edges, or because it strikes some intermediate balance between these extremes. The present paper is built around distinguishing among those possibilities.

2.3 Competing Organizational Possibilities

The central conceptual possibility explored here is that lower-tail organization may differ qualitatively even when lower-tail depth appears similar. We consider three broad possibilities.

The first is **sparse targeted recruitment**. Under this possibility, targeted shortcut placement broadens the role of the lower tail by routing a larger fraction of the system's total traffic through lower-tail edges. The tail then acts as a distributed load-bearing scaffold: not necessarily maximally exploited per edge, but broadly recruited into the transport economy of the graph.

The second is **random concentrated exploitation**. Under this possibility, random placement does not recruit the tail more broadly, but instead produces a smaller subset of tail edges that carry heavier burden relative to the rest of the graph. Here the lower tail acts less like a widely engaged scaffold and more like a hot exploited backbone (Serrano et al., 2009; Tadić et al., 2004). This possibility predicts that random placement may exceed targeted placement on metrics of per-edge tail exploitation even if targeted placement exceeds it on total tail participation.

The third is **saturation and washout**. As the number of active distal shortcuts increases, the distinction between selective recruitment and indiscriminate access may weaken (Watts and Strogatz, 1998). In that regime, participation differences may diminish even if burden concentration differences persist, and objective differences may become smaller or less interpretable. This possibility is especially important because it prevents the argument from collapsing into a simplistic "more tail is always better" narrative. Instead, it frames the problem as one of regime-dependent organization (Arenas et al., 2001).

Taken together, these competing possibilities define the logic of the paper. The study does not test whether lower-tail curvature matters in some generic sense. It tests whether the lower tail functions as a genuine routing object, and if so, whether its contribution to performance is better explained by broad recruitment, concentrated exploitation, or regime-dependent saturation effects. That is the specific mechanistic question opened by the earlier CAH control-law result (Pender, 2026b), and it is the question the present simulations are designed to answer.

3 Methodology

3.1 Base Graph Construction

All simulations were performed on synthetic hierarchical transport networks constructed as balanced trees. Each base graph was initialized as an undirected weighted tree with all basal edges assigned unit weight. Unless otherwise noted, the root node defined the reference point

for graph depth calculations, and node depth was computed as graph-theoretic distance from the root.

To explore the role of architecture, the base hierarchy was parameterized by branching factor and depth. The branching factor determined the number of children per non-leaf node, while depth determined the number of hierarchical layers in the tree. These two parameters jointly controlled the size, separation structure, and bottleneck geometry of the transport substrate. Because the aim of the present study was not to test arbitrary graph topologies, but to examine how lower-tail structure behaves in explicitly hierarchical networks, balanced trees served as the primary graph family throughout (Clauset et al., 2008; Dodds et al., 2003).

On top of each base hierarchy, a pool of candidate distal shortcut edges was generated. These candidate edges were defined as non-adjacent node pairs not already connected in the base tree and separated by a minimum graph distance ≥ 4 in the original hierarchy. In the present simulations, distal candidates were further restricted to pairs of nodes at the same depth, thereby favoring branch-bridging shortcuts that relieved hierarchical bottlenecks without collapsing the hierarchy into indiscriminate random connectivity (Kleinberg, 2001; Watts et al., 2002). Candidate generation therefore preserved the logic of the earlier CAH control-law construction (Pender, 2026b), in which distal access was intended to provide selective alternatives to root-dominated routing rather than a wholesale destruction of hierarchical structure.

3.2 Intervention Families

The core experimental manipulation compared two sparse shortcut-placement regimes applied to the same distal candidate pool: **targeted sparse placement** and **random sparse placement**. In both cases, the number of active distal edges was held fixed within a given condition, allowing transport differences to be attributed to edge organization rather than to total edge count.

In the **random** placement condition, active distal shortcuts were drawn uniformly from the candidate pool after random shuffling. This condition served as the control for whether sparse distal access alone was sufficient to alter lower-tail transport structure, independent of any deliberate placement logic (Watts and Strogatz, 1998).

In the **targeted** placement condition, active distal shortcuts were selected using a simple heuristic designed to favor high-separation branch-bridging edges. Candidate edges were scored according to their graph distance and node-depth profile in the base tree, and the highest-scoring edges were preferentially activated and assigned a uniform traversal weight (e.g. $w = 0.8$) to simulate high-conductance shortcuts. This operationalized a minimal notion of selective lower-tail organization: the targeted condition did not optimize transport globally through iterative search, but it did preferentially recruit shortcuts likely to alter routing structure in a systematic rather than arbitrary way (Kleinberg, 2000; Gastner and Newman, 2006).

Sparse edge budgets were examined at three levels:

- **n05**: 5 active distal edges
- **n10**: 10 active distal edges
- **n20**: 20 active distal edges

These budgets were chosen to probe distinct structural regimes. The n05 condition represents a highly sparse intervention regime in which each active shortcut has potentially large marginal influence. The n10 condition probes an intermediate regime in which selective organization and incipient saturation may compete. The n20 condition approaches a more shortcut-rich regime in which broad distal access may begin to wash out the distinctive effects of targeted organization (Barthélemy, 2011).

3.3 Transport Simulation

Transport was simulated by repeatedly sampling random source-target node pairs and routing them through the graph using weighted shortest paths (Dijkstra, 1959; Opsahl et al., 2010). For each sampled pair, the shortest path was computed with edge weights interpreted as traversal costs. The total cost of the path was then obtained by summing the weights of all edges along the route. This produced a per-pair transport cost and a per-pair hop count.

The transport simulation therefore yielded two classes of outputs. First, it generated pair-level observables such as path cost and path length, which could be summarized across many sampled pairs to estimate the graph's overall transport characteristics. Second, it generated edge-level traffic counts by incrementing an edge-load counter each time a sampled shortest path traversed that edge (Freeman, 1977; Brandes, 2001). These edge-load counts provided the basis for identifying whether lower-tail edges were genuinely load-bearing and for distinguishing broad lower-tail participation from concentrated tail exploitation.

For each condition, a fixed number of source-target pairs (N=1,000) was sampled using a seeded random number generator. This stochastic sampling procedure made it possible to estimate transport statistics repeatedly across seeds while preserving controlled comparisons between intervention families. Because the emphasis of the present study was not exhaustive all-pairs routing but repeated relative comparison across many graph instances and conditions, sampled transport was used as a computationally efficient proxy for the routing burden imposed by each intervention (Eppstein and Wang, 2001; Bader et al., 2007).

3.4 Curvature Metrics

Graph geometry was quantified using Ollivier–Ricci curvature (ORC) computed on edges (Ollivier, 2009; Lin et al., 2011). For each graph, ORC was estimated using a weighted implementation in which edge weights defined the graph metric and the transport kernels were parameterized by the standard ORC machinery. In practice, curvature was computed across all edges of the graph after the intervention-specific distal shortcuts had been added.

Particular emphasis was placed on the **lower tail** of the edge-curvature distribution rather than on the graph-wide mean. Following our CAH control-law extension (Pender, 2026b), the primary lower-tail summary statistic was q_{10}, defined as the 10th percentile of the edge-curvature distribution. Lower values of q_{10} correspond to deeper negative curvature in the lower tail. In addition to the mean and median curvature, the lower-tail quantiles were retained as separate summaries in order to test whether transport structure was better explained by lower-tail geometry than by average geometry alone.

To isolate the subset of edges constituting the lower-curvature tail in a given graph, edges were labeled as **tail edges** if their curvature value fell at or below the chosen lower-tail cutoff. In the present analyses, this was generally defined using the q_{10} threshold, so that the tail consisted of the most negatively curved 10% of edges. This tail-edge set was then used to compute load-bearing metrics such as tail load fraction, tail efficiency ratio, and tail burden concentration.

Because repeated simulations reused the same node labels across distinct graph realizations and conditions, curvature computation was performed with explicit cache clearing prior to each ORC run in order to prevent stale neighborhood-mass reuse across conditions. On Windows-based systems, a local multiprocessing compatibility workaround was also used to ensure ORC computation remained stable under the GraphRicciCurvature library (Ni et al., 2015, 2019).

3.5 Summary Metrics

For each graph condition, transport, curvature, and control observables were summarized into a flat result row suitable for repeated-seed aggregation and downstream comparison. The principal

summary metrics were as follows:

- **objective_J:** the composite control objective, combining transport cost, maintenance burden, congestion, and centralization penalties
- **mean_transport_cost:** the average weighted shortest-path cost across sampled source-target pairs
- **q10_curvature:** the 10th percentile of the edge-curvature distribution
- **tail_load_fraction:** the fraction of total sampled transport load borne by tail edges
- **tail_edge_fraction_loaded:** the fraction of loaded edges belonging to the tail
- **tail_efficiency_ratio:** tail load fraction normalized by tail edge fraction loaded
- **tail_burden_concentration:** mean tail-edge load divided by mean non-tail-edge load

The composite objective J was defined phenomenologically rather than derived from first principles. It was designed to retain continuity with the CAH control-law framework by penalizing high transport cost (Pender, 2026b), high maintenance burden, and concentrated bottlenecks while rewarding global routing efficiency only indirectly through the constituent graph metrics. Lower values of J indicate better overall performance under the assumed transport-maintenance tradeoff.

Three lower-tail burden metrics were especially important for interpretation. tail_load_fraction measures **total tail participation** and answers the question: how much of the system's total traffic runs through the lower-curvature tail? tail_efficiency_ratio measures whether the tail carries more than its fair share of traffic relative to how many loaded tail edges it occupies. tail_burden_concentration measures average per-edge burden inside the tail relative to the non-tail bulk. We distinguish between total tail participation, measured by tail_load_fraction, and per-edge tail exploitation, measured by both tail_efficiency_ratio and tail_burden_concentration, in order to separate **broad lower-tail recruitment** from **concentrated lower-tail burdening**.

Additional graph-wide summaries, including weighted mean shortest path, global efficiency, and edge betweenness statistics, were also retained to help interpret how intervention family and edge budget affected the broader routing landscape. However, the present study focused primarily on the relationship between lower-tail geometry and lower-tail load organization.

3.6 Repeated-seed Aggregation

To assess whether the observed effects were stable or idiosyncratic to a single stochastic realization, all primary experiments were repeated across multiple random seeds. Each seed determined the shuffle order of distal candidate pools, the random pair sampling used in transport simulation, and the condition-specific realizations of random sparse placement. Per-seed result tables were saved separately and later concatenated into a combined dataset for aggregate analysis.

For each condition, repeated-seed aggregation computed the mean, standard deviation, minimum, maximum, and count of the primary summary metrics. These aggregated summaries allowed us to distinguish robust directional effects from visually compelling but unstable single-run outcomes.

In addition to raw condition-level aggregation, we computed **paired targeted-versus-random differences** at matched edge budgets. For each seed and each edge budget (n05, n10, n20), the targeted value of a metric was subtracted from the corresponding random value or vice versa depending on the interpretive convention. In the present analyses, paired differences were primarily reported as:

- tail_load_fraction_diff = targeted - random
- objective_J_diff = targeted - random
- tail_efficiency_ratio_diff = targeted - random
- tail_burden_concentration_diff = targeted - random

This paired-difference analysis compressed the repeated-seed results into a more interpretable regime-level statement. Positive values of tail_load_fraction_diff indicate greater total tail participation under targeted placement. Negative values of objective_J_diff indicate better overall performance under targeted placement. Positive values of tail_efficiency_ratio_diff or tail_burden_concentration_diff indicate greater per-edge tail exploitation under targeted placement, whereas negative values favor random placement on those dimensions.

Together, the repeated-seed aggregation and paired-difference summaries were used to determine whether lower-tail organization was stable across seeds, whether targeted and random placement differed systematically in how they recruited and burdened the tail, and whether those differences translated into consistent objective advantages across sparsity regimes.

4 Results

4.1 The Lower Curvature Tail is a Load-bearing Transport Structure

The first question was whether the lower curvature tail marks a real routing object rather than a descriptive byproduct of the curvature calculation. The edge-level load-versus-curvature analyses indicate that it does. In both targeted and random sparse-placement conditions, lower-tail edges consistently carried disproportionate transport load relative to much of the remaining graph. This establishes the minimal claim of the paper: the lower curvature tail is not merely a summary of graph geometry, but corresponds to a functionally important routing substructure.

This result is consistent with the earlier CAH control-law extension finding that the optimum is tracked more reliably by the lower tail than by the graph-wide mean. However, it sharpens that earlier result by showing why lower-tail curvature is informative: it identifies the portion of the graph most directly implicated in routing burden. In this sense, the lower tail is not simply "the most negative part of the distribution." It is the part of the distribution where a disproportionate fraction of the network's transport work is concentrated.

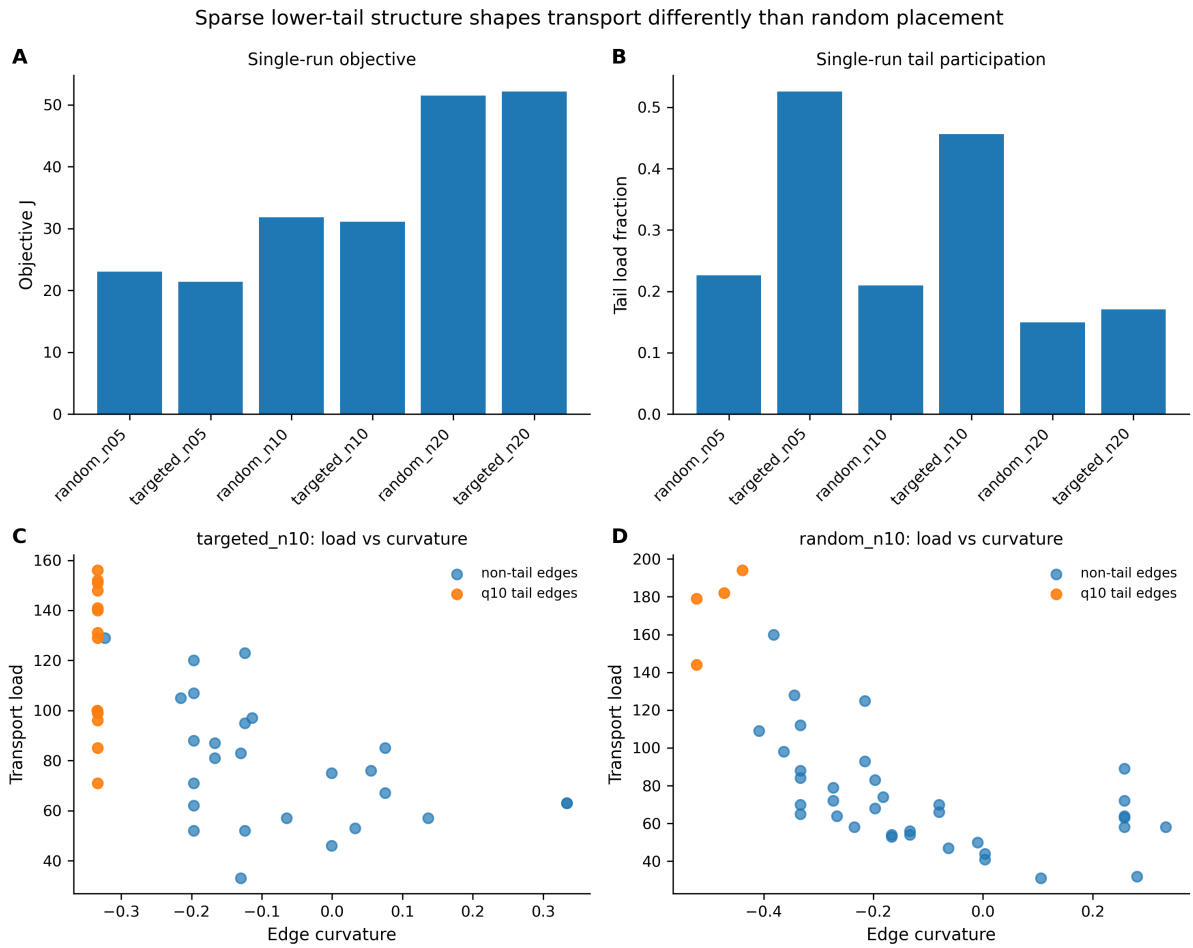

Figure 2: **Sparse lower-tail structure shapes transport differently than random placement. (A)** Single-run values of the composite objective J across targeted and random sparse-placement conditions at three edge budgets. **(B)** Single-run tail load fraction across the same conditions, showing that targeted placement can route a larger share of total traffic through the lower-curvature tail. **(C–D)** Edge-level transport load versus edge curvature for targeted_n10 and random_n10, respectively. In both cases, q_{10}-tail edges bear disproportionate transport load relative to much of the remaining graph, supporting the claim that the lower curvature tail is a genuine load-bearing routing structure. This figure is intended as a proof-of-concept visualization of the mechanism; the stability and regime dependence of these effects are evaluated in the repeated-seed analyses that follow.

4.2 Targeted Sparse Placement increases Total Tail Participation in the Sparse Regime

The repeated-seed aggregation shows that targeted placement most clearly increases total traffic routed through the lower tail in the sparse regime. The clearest effect appears at the lowest edge budget (n05), where targeted placement yields a stable increase in tail load fraction relative to matched random placement. At n10, the effect remains directionally positive but becomes substantially noisier. By n20, the separation weakens further, suggesting that the participation advantage of targeted placement is strongest before the shortcut structure approaches saturation.

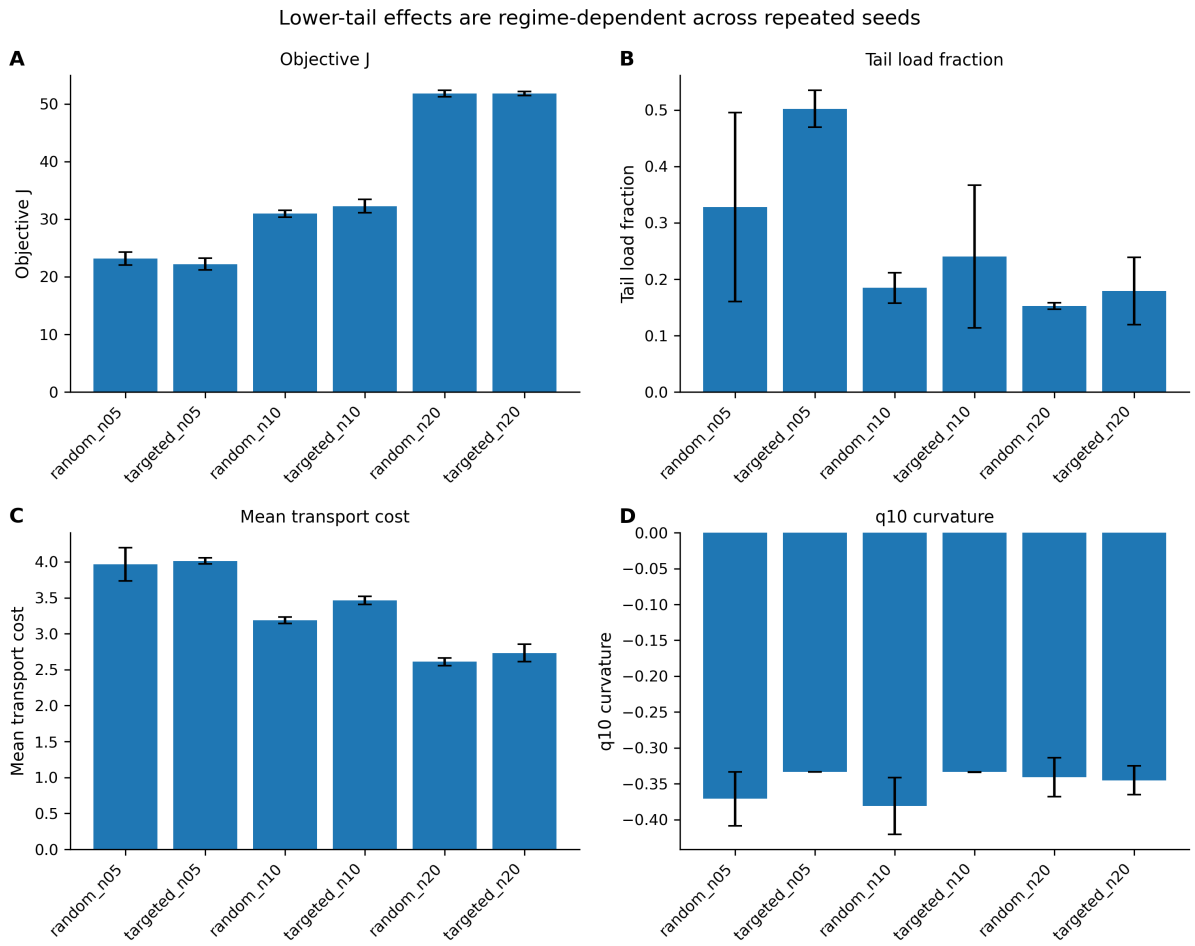

Figure 3: **Lower-tail effects are regime-dependent across repeated seeds. (A)** Aggregated composite objective J across targeted and random sparse-placement conditions at three edge budgets. **(B)** Aggregated tail load fraction, showing that targeted placement most clearly increases total lower-tail participation in the sparse regime (n05), with weaker and more variable separation at higher budgets. **(C)** Aggregated mean transport cost. **(D)** Aggregated lower-tail curvature q_{10}. Together, these repeated-seed results show that lower-tail effects are strongest and clearest in the sparse regime, while also indicating that lower-tail depth alone does not uniquely determine objective performance. The figure therefore supports the transition from simple lower-tail correlation to a more structured account of lower-tail organization.

This result is important because it identifies one specific mode of lower-tail organization: targeted sparse placement appears to recruit the lower tail more broadly into the routing substrate. In other words, targeted placement does not simply make the tail "more negative." It makes the tail carry more of the system's total traffic, at least in the sparse regime. This is the clearest positive result in favor of targeted organization.

4.3 Lower-tail Depth Alone does not Explain Objective Performance

Although lower-tail curvature remains more informative than mean curvature, the present results show that lower-tail depth alone does not explain transport performance. Targeted and random conditions can occupy similar or overlapping q_{10} regimes while differing in transport cost, traffic distribution, and objective value. This directly rules out the simplest scalar interpretation of the earlier CAH control-law extension result (Pender, 2026b).

The point is not that q_{10} is unimportant. On the contrary, the lower tail continues to identify the routing-relevant portion of the graph more effectively than mean curvature. But the repeated-seed analyses indicate that the same rough lower-tail depth can support different transport organizations. Lower-tail depth is therefore informative but not sufficient. The lower tail must be understood as a structured transport object rather than as a one-dimensional geometric coordinate.

4.4 Lower-tail Organization Decomposes into Recruitment and Concentration

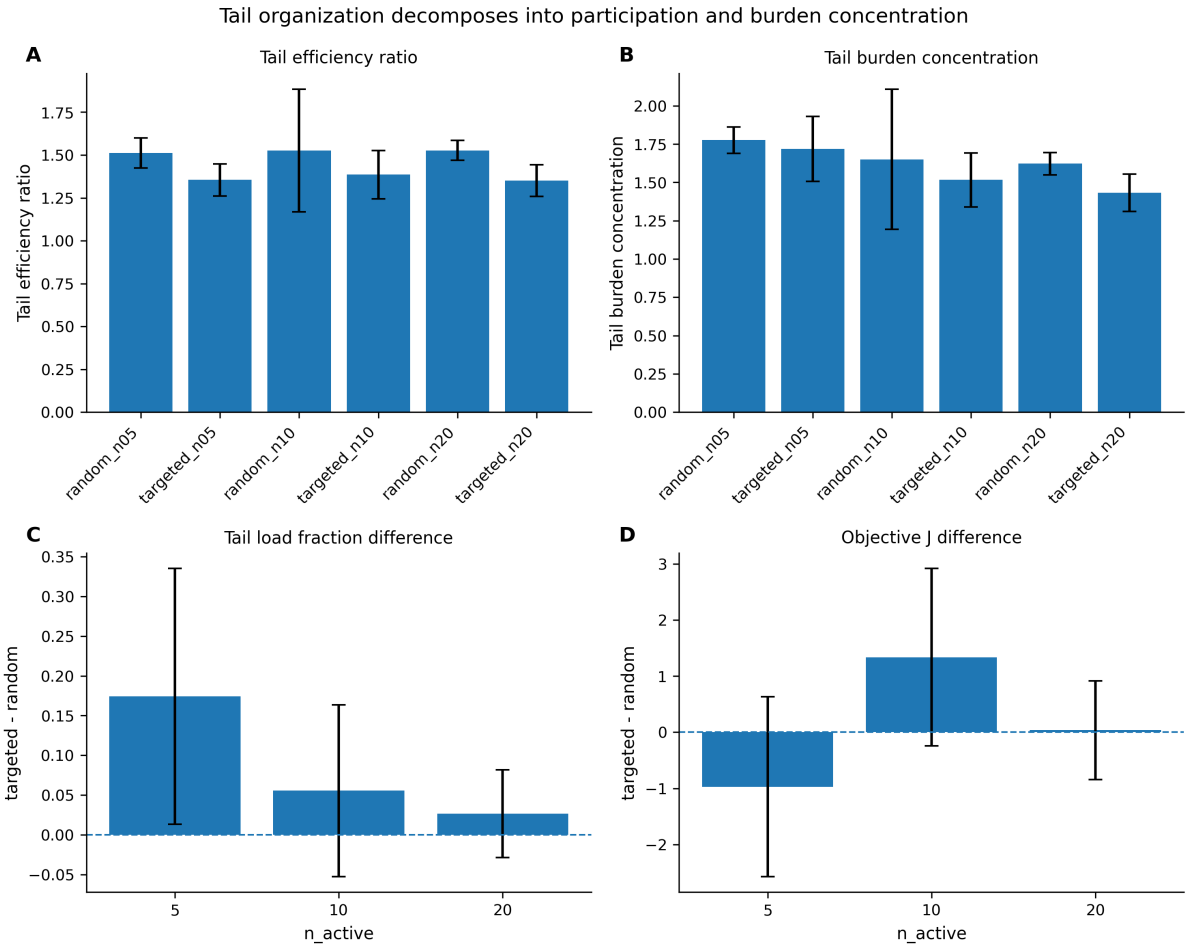

Figure 4: **Tail organization decomposes into participation and burden concentration.** **(A)** Aggregated tail efficiency ratio across targeted and random sparse-placement conditions at three edge budgets. **(B)** Aggregated tail burden concentration, showing that random placement more often produces heavier per-edge burden within the lower tail relative to the non-tail bulk. **(C)** Paired targeted-minus-random differences in tail load fraction, where positive values indicate broader lower-tail recruitment under targeted placement. **(D)** Paired targeted-minus-random differences in the composite objective J, showing that global performance remains mixed even when lower-tail organizational differences are present. Together, these panels support the paper's central mechanistic claim: lower-tail organization is not one-dimensional, but decomposes into broad recruitment versus concentrated exploitation, and global performance does not track any single tail metric in isolation.

The next step was to determine whether lower-tail organization itself could be decomposed into distinct modes. The answer appears to be yes. Targeted placement tends to increase total tail participation, but random placement more often yields greater per-edge exploitation intensity within the tail. This appears in two related metrics. First, tail efficiency ratio indicates that random placement often causes the tail to carry more than its proportional share of traffic relative to its loaded-edge footprint. Second, tail burden concentration shows that the average tail edge in the random condition tends to carry more load relative to the average non-tail edge than in the targeted condition.

Together, these results suggest that the lower tail is not organized along a single axis. At minimum, two distinct organizational dimensions are visible in the present data. One is broad recruitment of the lower tail as a routing substrate. The other is concentrated exploitation of the tail edges that do participate. Targeted placement tends toward the former; random placement tends toward the latter. This is the main mechanistic refinement to emerge from the study.

4.5 Global Performance Reflects a Tradeoff between Lower-tail Regimes

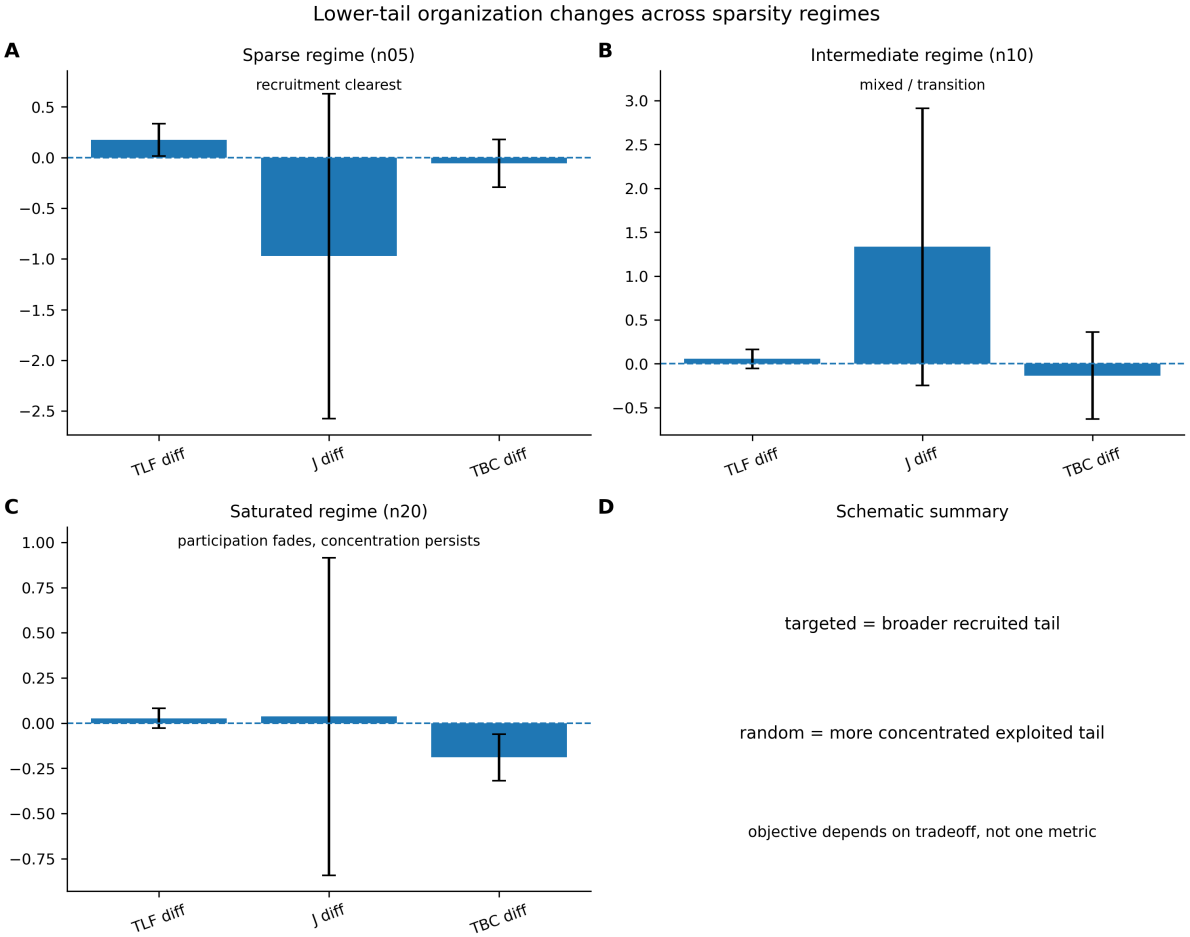

Figure 5: **Lower-tail organization changes across sparsity regimes. (A)** Sparse regime (n05), where targeted placement most clearly increases total lower-tail participation and the recruitment signal is strongest. **(B)** Intermediate regime (n10), which behaves as a noisier transition zone in which participation and concentration effects pull in different directions. **(C)** Saturated regime (n20), where participation differences weaken while burden-concentration differences persist more clearly. **(D)** Schematic summary of the resulting interpretation: targeted placement tends to produce broader recruited lower-tail structure, whereas random placement more often yields more concentrated exploited lower-tail structure. The figure summarizes the paper's regime-level conclusion that lower-tail organization, rather than lower-tail depth alone, governs the transport tradeoff.

The mixed behavior of the global objective J becomes more intelligible once these two modes are separated. Targeted placement does not universally outperform random placement on J. At n05, it can modestly improve the objective while also increasing total tail participation. At n10, the results become mixed, suggesting a noisier transition regime in which broader recruitment and hotter exploitation pull in different directions. By n20, objective differences are largely washed out even as burden concentration differences persist more clearly.

This pattern argues against any monotonic story. Neither broader recruitment nor more concentrated exploitation is automatically superior across all conditions. Instead, the global objective seems to depend on how those two modes trade off under a given sparsity regime. The sparse regime (n05) appears to favor broader recruitment; the intermediate regime (n10)

behaves more like a transition zone; and the more saturated regime (n20) reduces participation differences while leaving residual burden-concentration effects. The lower-tail scaffold is therefore not only real, but regime-structured.

4.6 Summary of the Mechanistic Refinement

Taken together, the results support a specific reinterpretation of our earlier CAH control-law finding. Lower-tail curvature remains more informative than mean curvature, but the explanatory object is not lower-tail depth in isolation. Instead, the transport optimum depends on how the lower tail is organized across at least three dimensions: depth, total participation, and per-edge burden concentration. The lower tail is thus best understood not as a scalar clue, but as an organized routing object whose structure changes across sparsity regimes.

5 Discussion

The present results refine the geometric-control picture established by Dynamic Curvature Adaptation and the CAH control-law extension. In the foundational CAH framework, geometry is treated as a dynamic thermodynamic resource: targeted gating purchases global routing efficiency by reshaping the topology through which signals propagate. Dynamic Curvature Adaptation argued that this kind of control can unlock a negative-curvature regime in which transport cost falls nonlinearly (Pender, 2026a), mitigating the burdens that flat Euclidean routing would otherwise impose. Our extension then recast that process as an explicit control problem and showed that the useful regime is not maximal hyperbolicity, but an intermediate optimum tracked better by lower-tail curvature than by mean curvature (Pender, 2026b).

The current study preserves those conclusions while refining their internal mechanism. The lower curvature tail is indeed a genuine transport object. The edge-level and repeated-seed analyses show that lower-tail edges consistently bear disproportionate traffic, which means that the lower tail is not simply a descriptive feature of the curvature distribution. This justifies the original control-law intuition that q_{10} captures something operationally relevant and supports treating the lower tail as a real routing substructure.

At the same time, the results argue strongly against a scalar interpretation of that substructure. No single tail metric is sufficient to explain performance. Lower-tail depth remains informative, but it does not by itself determine objective behavior. Tail load fraction identifies broad recruitment of the lower tail into the routing substrate. Tail efficiency ratio and tail burden concentration, by contrast, identify hotter exploitation of the tail edges that do participate. These organizational properties can diverge. A system may recruit the tail broadly without concentrating burden intensely, or concentrate burden intensely without maximizing total tail participation. The optimum therefore cannot be understood as a simple function of q_{10} or of any one tail metric in isolation.

This is the central conceptual shift of the paper. The relevant object is not the lower tail in isolation, but the organization of the lower tail across at least three partially separable dimensions: depth, total participation, and per-edge burden concentration. Under this interpretation, the control problem is not one of simply making curvature more negative, nor even of driving the lower tail to a particular q_{10} value. It is instead a problem of sculpting a lower-tail routing scaffold that is selective enough to reduce transport cost, distributed enough to avoid pathological overburdening, and sparse enough to avoid shortcut saturation. Lower-tail curvature is real, but its usefulness depends on how the tail is recruited and burdened.

This refined interpretation also clarifies the relationship to our CAH control-law extension. That earlier work identified the lower tail as the most informative geometric correlate of the optimum and showed that efficient transport requires preserving a selectively negative routing substructure without allowing the system to remain trapped in the deepest bottlenecked regime.

The present results suggest that our previous q_{10} finding was not the final answer, but the clue that led to a richer mechanism. The control law may not be governing lower-tail depth alone, but lower-tail organization. In this sense, the present study turns q_{10} from a useful correlate into an entry point for a more structured account of geometric control.

The regime structure visible across the three edge budgets reinforces this interpretation. In the sparse regime (n05), targeted placement most clearly broadens tail recruitment and can modestly improve the objective. In the intermediate regime (n10), participation and concentration effects pull in different directions, producing larger variance and more ambiguous objective outcomes. In the more saturated regime (n20), participation differences largely wash out while burden concentration differences remain visible. This suggests that placement effects are strongest when the lower-tail scaffold is still sparse enough to be special. Once too many distal routes are active, the system approaches a regime in which the distinction between selective recruitment and indiscriminate access begins to collapse.

These conclusions carry broader implications for the CAH program. In Dynamic Curvature Adaptation, targeted gating actively warped the network into a more efficient representational regime, and in our CAH extension that geometric modulation acquired the status of a control-law problem. The present work suggests that the deeper control target may be the organization of a lower-tail routing scaffold rather than curvature magnitude in the abstract. This makes the control story more mechanistic. It also provides a stronger middle layer between the foundational dynamic-curvature framework and more speculative extensions into hardware, learning rules, and cognition. Mechanisms analogous to SST-mediated gating in non-biological systems may need to regulate not just how negative curvature becomes, but how lower-tail recruitment and burden concentration are balanced under constraint.

Several limits remain important. The graphs are synthetic and hierarchical, the targeted placement heuristic is simple, and the objective J remains phenomenological rather than first-principles-derived. The current evidence is simulation-based rather than theorem-based, and it remains possible that different graph families or transport rules would alter the balance between recruitment and concentration. For this reason, the results should not be taken as proof of a universal theorem of lower-tail optimality. What they do provide is a clear mechanistic refinement: lower-tail curvature is more informative than mean curvature because it marks a real routing scaffold, but the contribution of that scaffold to performance depends on how it is organized.

The main conclusion is therefore straightforward. Mean curvature is too coarse, and lower-tail depth, while informative, is not the whole story. The lower curvature tail is a real, load-bearing routing structure, but it supports multiple distinct organizational strategies. Targeted placement broadens recruitment; random placement more often concentrates burden. The transport optimum emerges from the balance between those modes rather than from any single scalar geometric quantity.

6 Conclusion

This study began from a specific unresolved implication of the CAH control-law extension. If lower-tail curvature tracks transport optima more reliably than mean curvature, then the lower tail is unlikely to be merely a convenient summary statistic. The present results support that intuition and sharpen it substantially. Across repeated hierarchical transport simulations, lower-tail edges consistently bear disproportionate routing load, indicating that the lower curvature tail is a genuine load-bearing transport structure rather than a descriptive artifact of the curvature distribution. At the same time, the data show that lower-tail depth alone does not explain system performance. Conditions with similar q_{10} values can differ meaningfully in total tail participation, per-edge burden concentration, and global objective value, which means that

the lower tail cannot be reduced to a single scalar coordinate.

The main contribution of the paper is therefore a mechanistic refinement of the earlier control-law picture. The transport optimum is better understood not as a function of mean curvature, or even lower-tail depth alone, but as a tradeoff between distinct modes of lower-tail organization. Targeted sparse placement tends to broaden lower-tail recruitment, especially in the sparse regime, whereas random placement more often yields more concentrated exploitation of the tail edges that do participate. These effects do not map monotonically onto the global objective. Instead, they reveal that the lower tail supports multiple organizational strategies whose costs and benefits depend on sparsity regime and burden distribution. The explanatory object is not simply "negative curvature," but a selectively recruited and selectively burdened lower-tail routing scaffold.

This conclusion preserves the central insight of the CAH control-law extension while deepening its interpretation. That earlier work showed that the relevant operating point is an intermediate, condition-dependent optimum and that lower-tail curvature is a better geometric correlate of that optimum than the graph-wide mean. The present study suggests that this was not the final answer, but the clue that pointed toward a richer mechanism. The deeper control target may not be lower-tail depth in isolation, but the organization of the lower tail itself across depth, participation, and burden concentration. In this sense, the current work turns q_{10} from a useful correlate into an entry point for a more structured theory of geometric control in hierarchical transport networks.

More broadly, these findings strengthen the idea that controlled transport systems do not optimize by uniformly reducing geometric cost across the whole graph. They appear instead to benefit from preserving and regulating a specific lower-tail routing substructure that can be recruited, concentrated, or saturated depending on the control regime. That view is more mechanistic than a whole-graph curvature account and more honest than any blanket claim of lower-tail optimality. It suggests that future work in the broader CAH program, including extensions to adaptive hardware and geometry-aware learning rules, may need to regulate not only how negative curvature becomes, but how lower-tail routing burden is distributed under constraint.

The central takeaway is therefore straightforward. Mean curvature is too coarse, and lower-tail depth, while informative, is not the whole story. The lower curvature tail is a real, load-bearing routing structure, but its contribution to efficient transport depends on how it is organized. This reframes the lower tail from a scalar clue into an organized routing object and marks the clearest refinement of the CAH control-law story to date.

7 Data Availability

Code and simulation outputs supporting this study are available in the GitHub repository https://github.com/MPender08/beyond-mean-curvature ; an archived release corresponding to this manuscript version is available on Zenodo https://doi.org/10.5281/zenodo.19324655.

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